

Expense Tracker with AI Prediction

Project Report

Department of Computer Science and Engineering
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Project Title: Expense Tracker with AI Prediction

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Year: IV B.Tech

Abstract

Expense Tracker with AI Prediction is a web-based personal finance management system that helps users record income and expenses, visualize financial statistics, and predict monthly expenses using machine learning. The application provides a modern dashboard, transaction management, category-wise expense tracking, and AI-based expense estimation.

Technologies Used

- Python
- Flask
- HTML5
- CSS3
- JavaScript
- Bootstrap/Tailwind UI
- Pandas
- Scikit-learn
- NumPy
- Machine Learning
- CSV Dataset

Main Features

- Home Dashboard
- Add Income & Expense Transactions
- Category-wise Expense Tracking
- Financial Dashboard
- Expense Distribution
- AI Monthly Expense Prediction
- Responsive UI

- CSV-based Data Storage
- Machine Learning Prediction

Modules

Home Module

Overview of balance and navigation.

Transaction Module

Add and manage income/expense transactions.

Dashboard Module

Shows income, expenses, balance and charts.

Prediction Module

Predicts monthly expenses from income and spending inputs.

Database / Dataset

The project uses a CSV dataset (finance_data.csv) to train and test the machine learning model. Transactions can also be maintained for dashboard calculations.

Sample Screenshots

Image 1: Application interface demonstrating the Home page, Transaction form, Dashboard analytics, and AI Prediction module with a modern dark-themed responsive design.

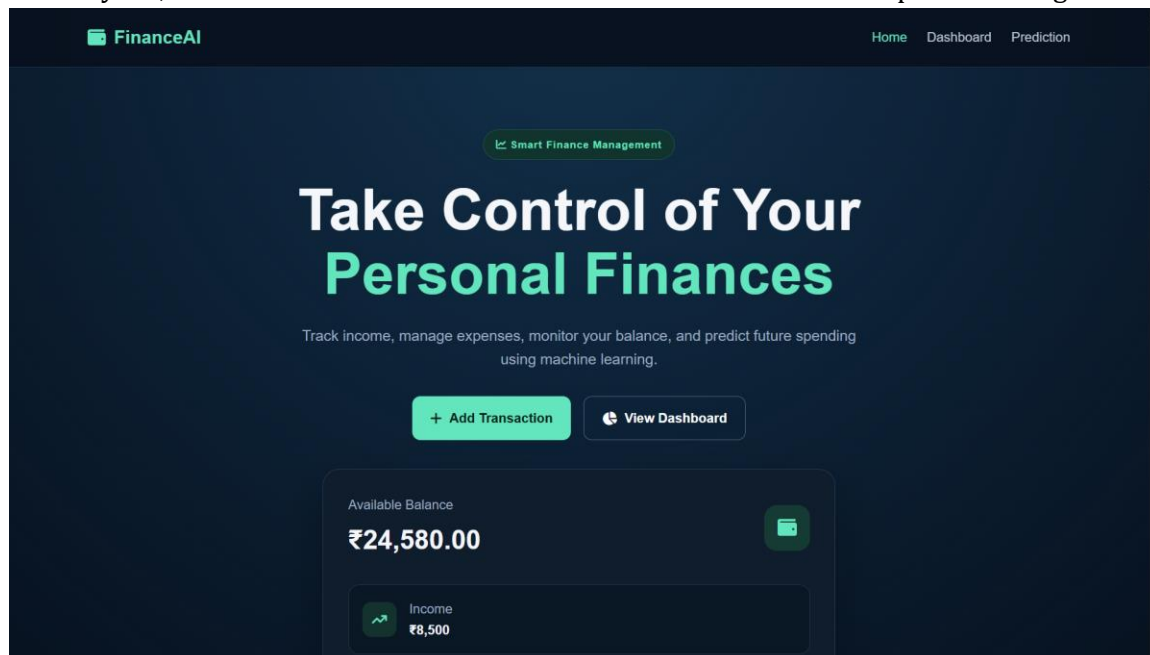


Image 2: Application interface demonstrating the Home page, Transaction form, Dashboard analytics, and AI Prediction module with a modern dark-themed responsive design.

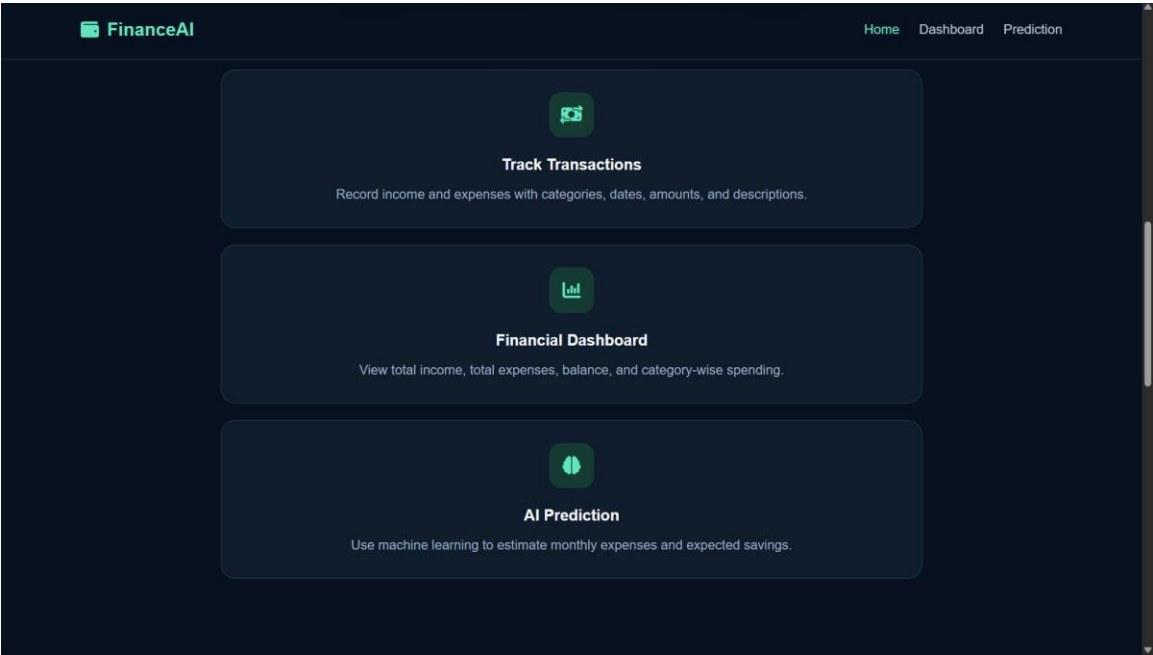


Image 3: Application interface demonstrating the Home page, Transaction form, Dashboard analytics, and AI Prediction module with a modern dark-themed responsive design.

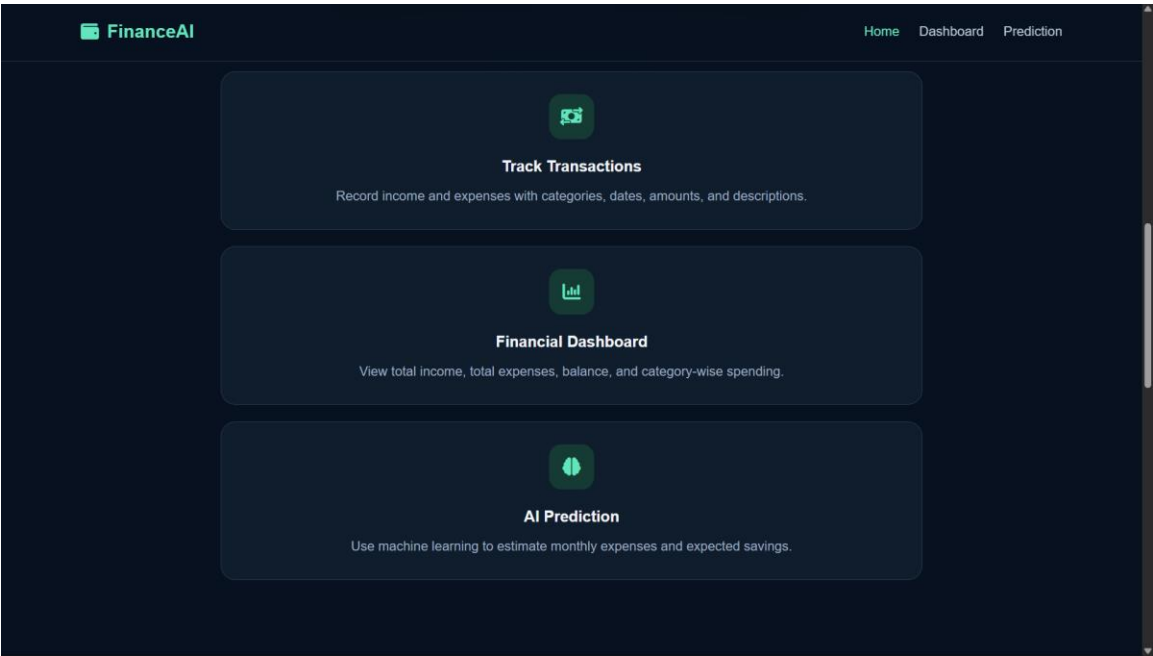


Image 4: Application interface demonstrating the Home page, Transaction form, Dashboard analytics, and AI Prediction module with a modern dark-themed responsive design.

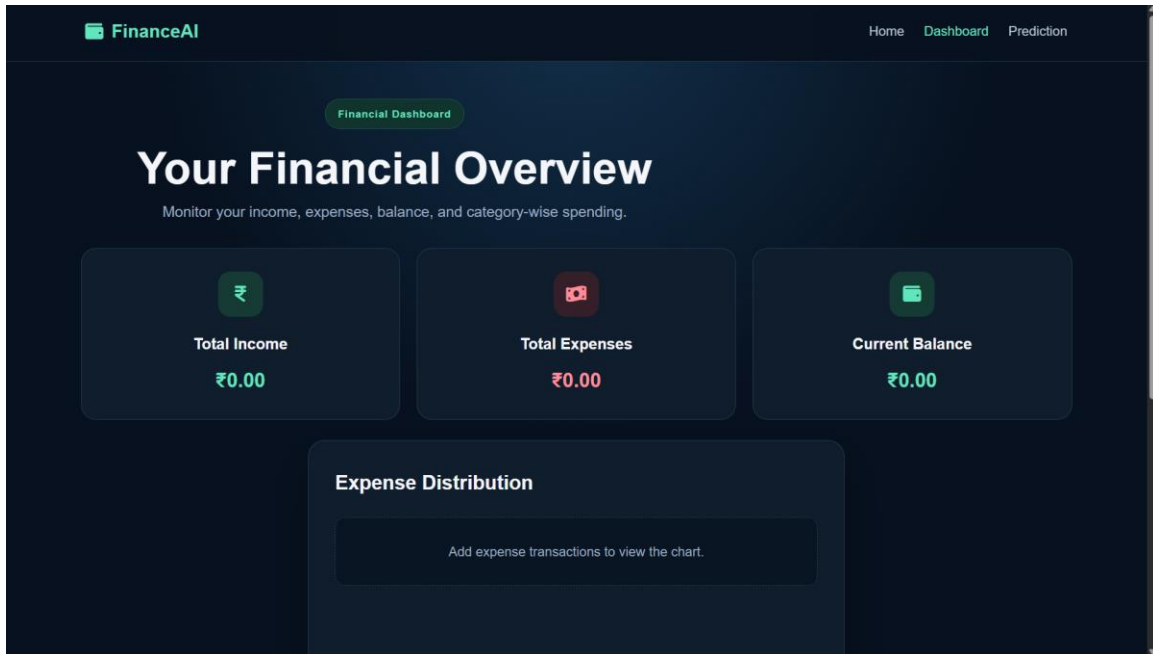
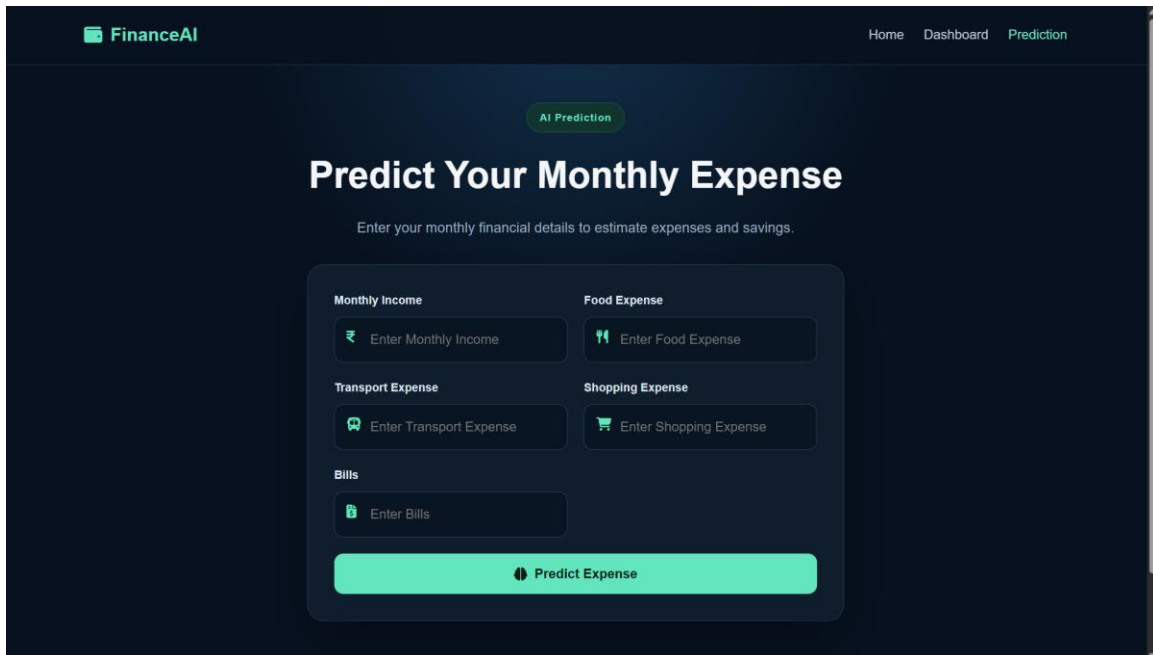


Image 5: Application interface demonstrating the Home page, Transaction form, Dashboard analytics, and AI Prediction module with a modern dark-themed responsive design.



Future Enhancements

- User Authentication
- Cloud Database
- Export Reports (PDF/Excel)
- Budget Alerts
- Email Notifications
- Mobile Application
- Multiple User Accounts

Commands to Run

```
python -m venv venv  
venv\Scripts\activate  
pip install -r requirements.txt  
python train_model.py  
python app.py
```